SIRAJ PATNAM – AI/ML Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience building scalable machine learning systems, NLP pipelines, and cloud-based production solutions across large-scale platforms, with strong expertise in recommendation systems, LLM-driven text understanding (RAG, GPT, LLaMA), PyTorch, Transformers, and Python-based microservices, and a proven track record of deploying real-time ML models, optimizing data pipelines for millions of users, and delivering high-performance AI solutions through end-to-end development, distributed training, API integration, CI/CD, and production monitoring.

EXPERIENCE

Perplexity AI - AI/ML Engineer **Jun 2024 – Present | CA**

- Designed and deployed LLM-powered AI systems for real-time question answering, semantic search, and conversational intelligence, leveraging state-of-the-art models such as GPT-based architectures, LLaMA, and open-source LLMs.
- Built end-to-end Retrieval-Augmented Generation (RAG) pipelines integrating vector databases (FAISS, Pinecone) with LLMs to deliver highly accurate, context-aware responses, significantly improving answer relevance and factual grounding.
- Fine-tuned and optimized large language models (LLMs) using techniques such as LoRA, PEFT, prompt engineering, and instruction tuning, improving domain-specific performance and reducing inference cost.
- Developed scalable NLP pipelines for text processing, summarization, entity recognition, and semantic similarity using Transformers (Hugging Face), SpaCy, and custom deep learning architectures.
- Implemented advanced prompt engineering and prompt chaining strategies, enabling better reasoning, multi-step problem solving, and improved response quality in generative AI applications.
- Built and managed MLOps pipelines for model training, evaluation, deployment, and monitoring using tools like MLflow, Docker, Kubernetes, Airflow, and CI/CD workflows, ensuring production-grade scalability and reliability.
- Designed low-latency inference systems using model optimization techniques such as quantization, distillation, batching, and caching, improving response time for real-time AI applications.
- Integrated vector search and embedding models (OpenAI embeddings, Sentence Transformers) to enable semantic retrieval, personalization, and ranking across large-scale unstructured datasets.
- Worked extensively with cloud platforms (AWS/GCP) for deploying scalable AI solutions, utilizing services like Lambda, S3, EC2, SageMaker, and Vertex AI.
- Built multi-agent AI systems and tool-augmented LLM workflows using frameworks like LangChain and LlamaIndex, enabling complex task automation and knowledge retrieval.
- Developed and maintained data pipelines for ingesting, cleaning, and transforming structured and unstructured data for LLM training and inference workflows.
- Collaborated cross-functionally with product and engineering teams to deliver AI-first features, improving user engagement and system intelligence across applications.
- Ensured responsible AI practices, including bias mitigation, hallucination reduction, and evaluation using automated metrics and human feedback loops.

Accenture – ML Engineer **Jun 2020 – July 2023 | India**

- Architected and deployed large-scale machine learning–driven ranking and recommendation systems, leveraging advanced Learning-to-Rank (LTR) techniques and Gradient Boosting models (XGBoost, LightGBM), significantly improving content relevance and user engagement across multiple platforms. Designed and optimized supervised and ensemble ML models including Logistic Regression, Random Forest, and XGBoost for high-impact use cases such as CTR prediction, spam detection, and feed ranking on high-dimensional, large-scale datasets.
- Built and trained advanced deep learning models using PyTorch and TensorFlow, including CNNs, LSTMs, and Transformer-based architectures (BERT, RoBERTa) for NLP tasks like text classification, semantic search, content understanding, and contextual ranking.
- Developed and fine-tuned state-of-the-art NLP models, applying techniques such as tokenization, embeddings, attention mechanisms, and transfer learning to improve semantic accuracy and language understanding in large-scale text data pipelines.
- Implemented embedding-based representation learning (Word2Vec, FastText, Sentence Transformers, custom embeddings) to enhance personalization, recommendation quality, and user-item similarity modeling.
- Performed advanced feature engineering and statistical analysis using Python libraries (Pandas, NumPy, Scikit-learn), deriving meaningful features from behavioral logs, graph-based signals, and engagement metrics to boost model performance.
- Applied robust model optimization techniques, including hyperparameter tuning (Grid Search, Random Search, Bayesian Optimization), regularization strategies, and cross-validation frameworks to improve generalization and minimize overfitting.
- Leveraged big data technologies such as Hadoop (HDFS, MapReduce), Spark, Hive, and Presto for distributed data processing, enabling efficient handling of terabyte-scale datasets and reducing model training and experimentation time.
- Designed and executed A/B testing and experimentation frameworks, applying statistical hypothesis testing to evaluate model performance, measure business impact, and drive data-backed decision-making.
- Collaborated on end-to-end ML lifecycle, including data ingestion, model development, validation, and deployment using containerized environments (Docker), ensuring scalable, low-latency, and production-ready systems.
- Maintained strong practices around experiment tracking, versioning, and reproducibility using tools like MLflow and Git, enabling faster iteration cycles and seamless cross-team collaboration.

SKILLS

- **Programming:** Python, SQL
- **Machine Learning & Recommendation Systems:** Supervised & Unsupervised Learning, Learning-to-Rank (Pairwise & Listwise), Collaborative Filtering, Candidate Generation & Ranking, Deep Learning (CNN, RNN, LSTM, GRU), Transformer Architectures, Time-Series Forecasting, Representation Learning, Hyperparameter Tuning (Grid, Random, Bayesian), Regularization Techniques, Model Evaluation (Precision@K, Recall@K, AUC, F1-score, ROC-AUC)
- **Generative AI & LLMs:** Prompt Engineering, Advanced Prompt Chaining, Retrieval-Augmented Generation (RAG), Semantic Search, LLM Fine-Tuning (LoRA, PEFT), Instruction Tuning, Embedding Models (Sentence Transformers, OpenAI), Structured Function Calling, Tool-augmented LLMs, Multi-Agent Systems, Hallucination Mitigation, LLM Evaluation (BLEU, ROUGE, Human Eval), Safety & Guardrails
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, Pandas, NumPy, SpaCy, NLTK, Matplotlib, Apache Spark
- **Data Engineering & Feature Systems:** ETL Pipelines, Data Pipelines for LLMs, Feature Engineering, Offline & Online Feature Stores, Training-Serving Parity, Data Preprocessing, Exploratory Data Analysis (EDA), Unstructured Data Processing
- **Cloud & Infrastructure:** AWS (S3, EC2, EKS, Lambda, SageMaker), GCP (Vertex AI, BigQuery), Docker, Kubernetes, Distributed Training (GPU/TPU), Mixed Precision Training, Data Parallelism, Model Quantization, Model Distillation
- **MLOps & Deployment:** MLflow, CI/CD for ML Pipelines, Model Deployment & Serving, A/B Testing & Experiment Design, Canary Releases, Model Monitoring, Data & Concept Drift Detection, Bias Detection, LLM Observability, REST APIs (Flask, FastAPI), Microservices Architecture
- **Databases & Retrieval:** Pinecone, FAISS, Vector Databases, Semantic Retrieval Systems, Hybrid Search (BM25 + Embeddings)
- **Methodologies:** Agile/Scrum, Cross-Functional Collaboration, Experimentation Frameworks, Data-Driven Decision Making

SKILLS

- **AWS Certified AI Practitioner – Amazon Web**
- **AWS Certified Machine Learning - Specialty – Amazon**
- **Google Advanced Data Analytics Specialization**
- **Accelerating End-to-End Data Science Workflows**

EDUCATION

Saint Louis University Saint Louis, USA

Master of Science in Computer Information Systems